

The EUMETSAT Network of Satellite Application Facilities  NWP SAF Numerical Weather Prediction	Cloud and Aerosol Detection Software Version 3	Doc ID : NWPSAF-EC-TV-031 Version : v1.4 Date : 15/02/2024
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Cloud and Aerosol Detection Software

Version 3

Test Plan

Version 1.4
15/02/2024

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ECMWF

This documentation was developed within the context of the EUMETSAT Satellite Application Facility on Numerical Weather Prediction (NWP SAF), under the Cooperation Agreement dated 7 December 2016, between EUMETSAT and the Met Office, UK, by one or more partners within the NWP SAF. The partners in the NWP SAF are the Met Office, ECMWF, DWD and Météo France.

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Change record			
Version	Date	Author / changed by	Remarks
1.0	01/04/2020	Reima Eresmaa	Initial version
1.1	28/04/2020	Reima Eresmaa	Include ECMWF test results
1.2	02/06/2020	Reima Eresmaa	Include a summary from beta testers' findings
1.3	07/07/2020	Reima Eresmaa	Clarify on the module testing and creation of test input and reference output datasets
1.4	15/02/2024	Chris Burrows	Update to cover additional functionality in CADS v3.2

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1 MODULE TESTS

Modules will first be compiled in isolation of each other and it will be checked that no warning (or error) messages are issued. The module testing will be carried out by the software developer on an Atos supercomputer using Fortran compilers of Nvidia, Intel, and GNU.

Output of the module testing will be recorded in the test folder.

2 INTEGRATION TESTS

The code will be compiled as a whole using a makefile and it will be required that no warning (or error) messages are issued. This will be verified by the software developer on a variety of computing platforms and Fortran compilers (`gfortran` and `ifort` and `nvidia` on an Atos supercomputer). The nominated beta testers will carry out the integration testing on their local computing platforms and Fortran compilers available to them. The beta testers will also review the accuracy and completeness of software documentation (including the release note and user guide).

Output of the integration testing will be recorded in the test folder all email exchange between the software developer and the nominated beta testers will be archived here.

3 VALIDATION TESTS

It will be checked that the software produces sensible output from different input datasets including real radiance data from the Atmospheric Infrared Sounder (AIRS), Infrared Atmospheric Sounding Interferometer (IASI), Cross-track Infrared Sounder (CrIS), Hyperspectral Infrared Atmospheric Sounder (HIRAS), Hyperspectral Infrared Atmospheric Sounder-2 (HIRAS-2), Geostationary Interferometric Infrared Sounder (GIIRS), Infrared Fourier Spectrometer -2 (IKFS-2) and Infra-Red Interferometer Spectrometer (IRIS), as well as simulated radiance data from the Meteosat Third Generation Infra Red Sounder (MTG-IRS) and Infrared Atmospheric Sounding Interferometer – Next Generation (IASI-NG). For this purpose, reference test runs (shell scripts), test input, and associated reference output datasets will be prepared. The test input and reference output datasets will be collected from the archived output of the screening process as implemented in the ECMWF NWP system. These datasets will also be supplied as part of the software package for users.

Processing time per observation will be determined in terms of consumed CPU time and this will be compared with time consumed in RTTOV forward model computation. It will be required that the CPU time consumed in cloud and aerosol flagging is small in comparison with that consumed in the forward modeling.

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Consistent performance with the ECMWF 4D-Var data assimilation system is required and this will be verified by detailed comparison of output cloud, aerosol, trace gas, and land-sensitivity flags against those produced inside the 4D-Var. These flags are required to be identical in samples consisting of several thousands of observations. This verification is done with all those IR sounders that have thus far provided us with real measurement data. The test samples will be made representative of global data such that percentages of cloud-, aerosol-, trace-gas-, and land-affected data closely match those in the ECMWF 4D-Var. To ensure this, information of the data rejection percentages is collected from ECMWF 4D-Var log files and the test scripts are written such that these percentages are included in the output.

Output of the validation testing will be recorded in the test folder.

4 PORTABILITY TESTS

As part of the final software package, test scripts will be provided for users to quickly verify that the software runs as expected and produces the desired output from given input data. The portability testing (to be carried out prior to the delivery phase) will rely on these test scripts. Emphasis will be put on ensuring that the software produces exactly the desired output regardless of which computing platform (Atos supercomputer or MacBook) or Fortran compiler (gfortran, nvidia, ifort) is used. Variation in computing time between the different compilers and platforms will be allowed.

Output of the portability testing will be recorded in the test folder.

5 TRACEABILITY MATRIX

Table I shows the traceability matrix and identifies the testing methods applied to each of the product requirements specified in the Product Specification document.

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Table I: Traceability matrix.

Requirement	Testing method	Test plan reference	Comment
1. The software compiles, runs and produces similar output on a variety of Unix platforms	Compiling by makefile and running test scripts on several platforms.	2,4	Beta testers' role will be essential
2. The software includes test scripts and sample data representative of AIRS, IASI, CrIS, HIRAS, HIRAS-2, GIIRS, IKFS-2, MTG-IRS, IASI-NG and IRIS sounders	Visual inspection of the files and directory structure as contained in the software package.	2	Beta testers' role will be essential
3. The software identifies cloud-contaminated IR channels at each location	Visual inspection of the output. Require that cloud flag is raised as often as in ECMWF 4D-Var.	3	
4. The software identifies locations where aerosol contamination occurs and distinguishes between Saharan dust, volcanic ash, and other aerosol	Visual inspection of the output. Require that aerosol type is included in output, and that all possible values (0, 1, 2, 3) are present.	3	
5. Depending on aerosol species, the software identifies contaminated IR channels at each location	Visual inspection of the output. Require that aerosol flag is raised as often as in ECMWF 4D-Var.	3	
6. The software identifies locations and IR channels affected by anomalously high concentration of atmospheric HCN	Visual inspection of the output. Require that trace-gas flag is raised as often as in ECMWF 4D-Var.	3	
7. The software identifies IR channels sensitive to underlying surface at each location	Visual inspection of the output. Require that land-sensitivity is flagged as often as in ECMWF 4D-Var.	3	
8. Production of the cloud, aerosol, trace-gas, and land-sensitivity flags is efficient and consumes only a small fraction of computing time as compared with forward modelling of brightness temperature	Using the Fortran built-in subroutine CPU_Time, determine time consumed per observation at each test script. Compare with computing times needed to run RTTOV.	3	
9. Perform consistently with the ECMWF 4D-Var system	One-to-one comparison of output flags with those produced by the 4D-Var.	3	

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6 RESULTS

6.1 Module testing

Individual source code files were compiled in isolation of each other on an Atos GPIL login node (AMD EPYC Rome, 2.5GHz) at ECMWF. The compiling option “-c” was applied. The module testing on the final setup on 15/02/2024 was successful on all tested Fortran compilers. No error or warning messages were seen. The full output of the final module testing is saved in file `module_testing.txt` in the test folder, and includes the exact compiler versions used in the testing.

6.2 Integration testing

The software was compiled as a whole using the makefile included in the source code directory. The compilation was repeated a number of times, each time starting from scratch, and modifying the makefile header according to the Fortran compiler in use. The same compilers and platforms as detailed above for the module testing were used in the integration testing. Specific action had to be taken to load/unload the ECMWF-specific module eccodes depending on which Fortran compiler -containing modules were in use at the same time. This is because of the incompatibility of certain external modules with the internal eccodes module. There were no error or warning messages in the output of the makefile runs. The full output of these tests is in the test folder (see file `integration_testing.txt`).

Both the release note and user guide documents have been modified to account for the new developments in the software.

6.3 Validation testing

The sensibility of the software output was verified by producing reference output files from the ECMWF 4D-Var system and comparing the software output against these. In all test samples that contain real measurement and observation minus model background data (i.e., those of AIRS, CrIS, GIIRS, HIRAS, IASI, and IKFS-2), the output flagging is found to be identical to the reference. Each sample consists of more than 1,000 observations and includes a selection representative of global sampling. The output flags from simulated IASI-NG, MTG-IRS, HIRAS-2 and IRIS is broadly consistent with that of the other supported IR sounders. In agreement with the Product Specification, aerosol and trace-gas detection is inactive for GIIRS, HIRAS, IKFS-2, IASI-NG, IRS, HIRAS-2 and IRIS, so that these flags are always set to zero.

The software’s computational efficiency was verified by determining the time it takes to process one observation from each of the supported sensors. The test data samples were used and the test script was rerun 100 times for each sensor to get a robust estimate of the processing time. A reference test environment using RTTOV V12 was created and computing time taken in then radiative transfer modelling was determined from there for comparison. The result is that, depending on the sounder, running RTTOV takes

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approximately 4-6 times more time than running CADS. The variation between different sounders is mainly due to different numbers of channels processed from each.

It is verified that the CADS output file contains the aerosol type information.

The detailed output of validation testing is saved in file `validation_testing.txt` in the test folder.

6.4 Portability testing

The software was installed, and the test script was run through with all the provided test input datasets on a login node of the Atos supercomputer, and a MacBook. All available Fortran compilers were used. In the final portability testing on 15/02/2024, the output flagging was in all tests identical to reference.

The testing on the MacBook revealed a bug whereby the array `N__Num_Regression` in `CADS_Setup_Aerosol.F90` was not initialised, and the undefined elements were used to initialise new arrays which resulted in memory limits being exceeded. This is thought to have been overlooked in the past due to some compilers initialising arrays to zero automatically.

After fixing these errors, all of module, integration and portability testing was successfully repeated, as was the validation testing as far as the identity of output flagging with reference was concerned.

File `portability_testing.txt` in the test folder contains detailed output of the portability testing on the ATOS system, and `portability_testing_macbook.txt` contains the MacBook output.